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ABSTRACT

This project presents the design and implementation of a Voice-Controlled IoT Robot using an ESP32 and a Bluetooth-enabled mobile application. The robot is controlled through voice commands from a smartphone, which are converted into movement instructions and transmitted via Bluetooth. The ESP32 processes these commands to control the robot's motion through a motor driver. Additionally, the system provides IoT-based feedback such as obstacle detection, movement status, and battery level for real-time monitoring. The solution is low-cost, efficient, and scalable, making it suitable for applications in automation, smart assistance, surveillance, and educational robotics.

1.INTRODUCTION

Robotics and Internet of Things (IoT) technologies are rapidly transforming the way machines interact with humans and their environment. The integration of voice recognition, wireless communication, and real-time data monitoring has led to the development of intelligent and user-friendly systems. A Voice-Controlled IoT Robot is one such innovation that enables users to control robotic movements using simple voice commands through a smartphone. This reduces the need for manual operation and enhances accessibility and efficiency.

By combining the capabilities of the ESP32 microcontroller with sensors and IoT platforms, the system not only performs actions based on user input but also provides real-time feedback. This makes the robot more reliable, interactive, and suitable for modern applications such as automation, surveillance, and smart assistance.

1.1 PURPOSE

The purpose of this project is to develop a Voice-Controlled IoT Robot that can understand and execute human voice commands. The system aims to provide an interactive, hands-free, and efficient solution for controlling robotic movements using modern IoT technologies.

1.2 SCOPE

This project involves the design of a robot using an ESP32 microcontroller with Bluetooth and Wi-Fi capabilities. It supports voice-based control and integrates IoT features for real-time monitoring. The system includes sensors for obstacle detection and battery monitoring, making it suitable for applications such as surveillance, automation, assistive robotics, and smart transportation.

1.3 DEFINITIONS, ABBREVIATIONS and MODEL DIAGRAM

- **IoT (Internet of Things):** A network of interconnected devices that communicate and share data.
- **ESP32:** A microcontroller with built-in Wi-Fi and Bluetooth used for IoT applications.
- **IR Sensor:** A sensor used for detecting obstacles.
- **Voltage Sensor:** Measures battery level of the system.
- **Motor Driver:** Controls the movement of motors based on commands.
- **Buzzer:** Provides alert signals for obstacles or low battery.

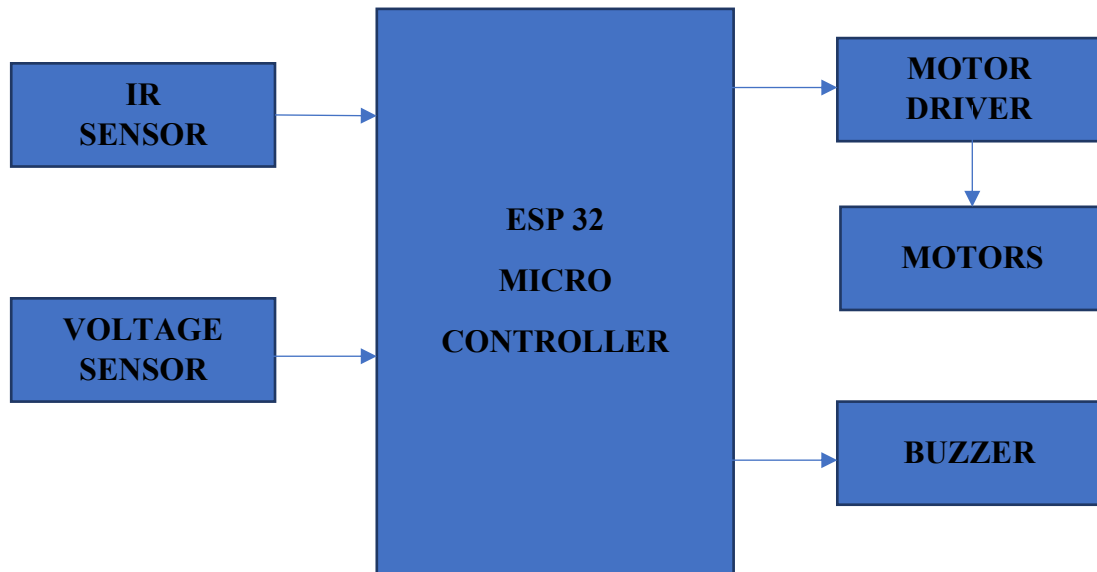


Figure 1.1: Model Diagram

1.4 OVERVIEW

Robotics and IoT technologies enable machines to interact intelligently with humans and their environment. In this system, voice commands are sent from a mobile application to the ESP32 via Bluetooth or cloud connectivity. The ESP32 processes these commands and controls the robot's movement accordingly.

The robot uses IR sensors for obstacle detection and voltage sensors for battery monitoring, ensuring safe and efficient operation. IoT integration allows real-time feedback such as movement status, obstacle alerts, and battery condition to be sent to the user. This enhances system reliability and makes it suitable for various modern automation applications.

2. LITERATURE SURVEY

“Voice Controlled Pick &Place Robo Using Android Application”

PradeepKumar, HariKumar.

Robotics has become an interesting area of research in the field of Electrical Engineering. This is due to the vast potential it holds in reducing human effort in performing tasks faster while still maintaining operational accuracy. The recently introduced Android operating system used in many devices such as mobile phones, tablets is user friendly and can be extended to control smart and intelligent systems such as robots. Our design develops a pick and place robotic arm vehicle using an android application to control the robot through voice commands. Arduino programming language, which is an open source and simplified version of C++ known for its friendly architecture was used to program the controller. The integration of control unit with blue-tooth device was done to capture and read the voice commands. The robotic vehicle then operates as per the command received via android application.

“Monitoring system in industry using IoT”

M Jagadesh, M Saravanan, V Narayanan, M Priya Vadhana, K Logeshwaran.

In today's web world IoT innovation has been definitely expanded. IoT is utilized to associate is a mix of correspondence framework and inserted framework. IoT is utilized for transmitting and accepting information. This frameworks are utilized to screen mechanical machines. In this procedure little scale modern applications like fluid dimension control, vitality checking and so on are observed remotely by mobiles and workstations. The fundamental thought of this paper is to outline significance of IoT to screen mechanical applications.

“Voice Controlled Personal Assistant Using Raspberry Pi”

Emad S. Othman.

This paper shows the working of a device based on implementation of a voice command system as an intelligent personal assistant. The services provided by the device depends on the input given in the form of voice command by the user and ability to access information from a variety of online sources such as weather, telling time or accessing online applications to listen to music. This Voice driven device uses Raspberry Pi as its main hardware. Speech to text engine is used to convert the voice command to simple text. Query processing is then applied using natural language processing (NLP) onto this text to interpret the intended meaning of the command given by the user. After interpreting the intended meaning, text to speech conversion is used to give appropriate output in the form of speech. This device might provide a platform to visually impair to do their day to day tasks more easily like listening to music, checking weather conditions, checking current time or even doing a simple mathematical calculation. Many experiments and results were accomplished and documented.

"Mobile Quad-Controlled wireless Robotic Arm,"

N. Reganit, S. Anantha Alli, B. C. V. Mohan, R. Mounica, and V.K. Mittal.

In this paper, we develop a remotely controlled robotic arm with 4 degree of freedom (D.O.F) that is wirelessly controlled using four control mechanisms, i.e, Voice Control, Smart Phone-Tilt Control, Remote control and Hand Gesture Control. Wireless technologies such as Bluetooth and Wi-Fi are used to access the Quad-Controlled Robotic Arm (QCRA). A prototype QCRA is developed. The QCRA can be used to pick and place objects from one place to another on receiving the commands from distance, thereby reducing the human effort. An Android application is developed for convenience of the user in operating this QCRA, using different control mechanisms. Performance evaluation results are encouraging. Potential applications of the QCRA in homes, industries and for physically challenged/aged people are also discussed.

"An Efficient Polynomial Pool-Based Scheme for Distributed Heterogeneous WSNs"

M. Senthil Kumar and M. Gopinath.

This paper proposes using a mobile collector, such as an airplane or a vehicle, to collect sensor data from remote fields. We present three different schedules for the collector: round-robin, rate-based, and min movement. The data are not immediately transmitted to the base station after being sensed but buffered at a cluster head; hence, it is important to ensure the latency is within an acceptable range. We compare the latency and the energy expended of the three schedules. We use the ns-2 network simulator to study the scenarios and illustrate conditions under which rate-based outperforms round-robin in latency, and vice-versa. The benefit of min movement is in minimizing the energy expended.

“An Assuring Approach for Tree-Based Routing Topology in WSNs”

M. Senthil Kumar and L. Praveen.

We investigate the following fundamental question-how fast can information be collected from a wireless sensor network organized as tree? To address this, we explore and evaluate a number of different techniques using realistic simulation models under the many-to-one communication paradigm known as convergecast. We first consider time scheduling on a single frequency channel with the aim of minimizing the number of time slots required (schedule length) to complete a convergecast. Next, we combine scheduling with transmission power control to mitigate the effects of interference, and show that while power control helps in reducing the schedule length under a single frequency, scheduling transmissions using multiple frequencies is more efficient. We give lower bounds on the schedule length when interference is completely eliminated, and propose algorithms that achieve these bounds. We also evaluate the performance of various channel assignment methods and find empirically that for moderate size networks of about 100 nodes, the use of multifrequency scheduling can suffice to eliminate most of the interference. Then, the data collection rate no longer remains limited by interference but by the topology of the routing tree. To this end, we construct degree-constrained spanning trees and capacitated minimal spanning trees, and show significant improvement in scheduling performance over different deployment densities. Lastly, we evaluate the impact of different interference and channel models on the schedule length.

“Network Selection in Heterogeneous Wireless Systems using GRA Method”

Senthilkumar Meyyappan and N. Selvamuthukumaran.

In heterogeneous wireless networks, connection choosing relates to the method for automatically determining the most suitable one among a variety of readily accessible alternatives according to features including quality of signal, connectivity, and preferences of users. The network choice with heterogeneous wireless systems includes the procedure for choosing a suitable system from an array of available alternatives, taking into consideration variables that include signal quality, bandwidth, and customer preferences, in order to guarantee ideal interaction as well as efficiency for wireless gadgets in a variety of environments. This is a dynamic choice that seeks to maximize the user experience by replying to shifting network situations and demands. Network selection study for heterogeneous wireless systems is crucial since it boosts wireless transmission by means of sophisticated algorithms, enhancing effectiveness, dependability, and overall performance while tackling obstacles from diverse technologies alongside rising connectivity demands, thus improving consumers along with allowing innovations in the Internet of Things and 5G wireless networks. The GRA (Grey Relational Analysis) is a decision-making tool which assesses the value of many variables in a grey structure, helping in the selection of the most appropriate alternative. In accordance with grey system theory, method examines reference and comparison sequences employing grey relational parameters, permitting impartial rankings especially supporting difficult decisions in the fields of engineering, leadership, and economics. The Evaluation parameters are Bandwidth, Jitter, Delay and Cost. Alternate parameter is Wi-Fi. For the analysis, the final rank is where “Wi-Fi 1” is in the 1st rank, “Wi-Fi 3” is in the 2nd rank, “Wi-Fi 4” is in the 3rd rank, “Wi-Fi 2” is in the 4th rank, and “Wi-Fi 5” is in the 5th rank. The best one is “Wi-Fi 1” which holds up the 1st rank.

“Secure Routing in MANETs with Key Management”

M. Kavitha, T. Maheshwaran and M. Senthil Kumar.

This paper presents a comprehensive, cost-effective, and intelligent surveillance system utilizing the AI Thinker ESP32-CAM module, integrating real-time video streaming, human detection, and audible alerts for enhanced security. Designed for applications such as home monitoring, industrial safety, and especially border surveillance, the system leverages a PIR motion sensor to trigger video capture and streaming via a local network. With video frames processed through a locally hosted Flask server using YOLO and Open CV for object detection, the presence of a person activates a buzzer and enables instant alerts through IoT connectivity. By combining low-cost hardware, open-source software, and deep learning techniques, including convolutional neural networks for robust performance in low-light and complex scenarios, the project offers a scalable and autonomous security solution. Specifically addressing India's extensive 15,106 km border with its neighboring countries, this system demonstrates significant potential for securing restricted and high-risk areas, effectively merging AI and IoT to meet modern surveillance demands.

“Unlocking Mobile Devices using Improved Face Recognition and Eye Blinking Technique”

K. Arutselvan, C. Sridhathan and M. Senthil Kumar.

In recent days, face recognition with eye blink counting techniques are very popularly used in smart phone platform for unlocking the devices. In conventional face recognition system some drawbacks are there such as photograph image of a original face can be used for unlocking the device. So it leads to insecure operation and leak of confidential information without the owner's knowledge. In this paper, face detection with eye blink counting based recognition methods for unlocking the mobile devices is discussed.

“Ensure Data Transmission in Mobile Ad-Hoc Networks”

M. Kavitha, T. Maheshwaran and M. Senthil Kumar.

In the existing faith management scheme, the faith model has two components: faith from direct observance and faith from indirect observance. With direct observation from an observer node, the trust value is derived using Bayesian inference, which is a type of unsure conclusion when the full probability model can be defined. On the other hand, with indirect observance, also called secondhand information that is obtained from neighbor nodes of the observer node, the trust value is derived using the Dempster-Shafer concept, which is another type of uncertain reasoning when the proposition of interest can be derived by an indirect method. We proposed DES concept to encrypt the data between source and destination. In this method used for securely transmit the data and achieve the secure data transmission in mobile ad hoc network.

2.1 TECHNOLOGIES USED

1. Microcontroller–ESP32

The ESP32 is the central processing unit and the most critical component of the voice-controlled IoT robot, acting as the brain that coordinates all operations. It is responsible for receiving, processing, and executing commands sent from the mobile application through Bluetooth communication. The ESP32 interprets these commands and generates appropriate control signals to drive the motors and other connected components, thereby enabling the robot to perform actions such as moving forward, backward, or turning in different directions.

One of the key advantages of the ESP32 is its powerful dual-core processor, which allows it to handle multiple tasks simultaneously, such as communication, sensor data processing, and motor control. It operates with high efficiency while consuming relatively low power, making it suitable for battery-operated systems. Additionally, the ESP32 has built-in Wi-Fi and Bluetooth capabilities, which eliminate the need for separate communication modules, thereby reducing system complexity, cost, and power consumption.

The microcontroller also supports various General Purpose Input/Output (GPIO) pins, Analog-to-Digital Converters (ADC), Pulse Width Modulation (PWM), and multiple communication protocols such as UART, SPI, and I2C. These features make it highly versatile and capable of interfacing with a wide range of sensors and actuators. Due to its flexibility, scalability, and performance, the ESP32 is widely used in IoT-based smart systems and robotics applications.

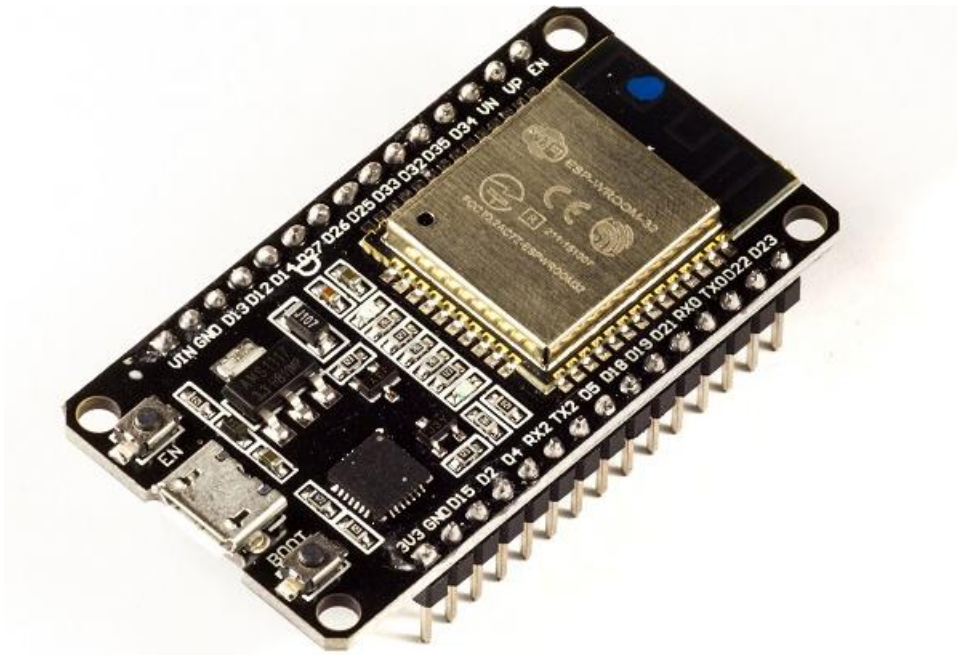


Figure 2.1: ESP32 Microcontroller

2. Programming Platform – Arduino IDE

The Arduino Integrated Development Environment (IDE) serves as the primary software platform for developing and uploading programs to the ESP32 microcontroller. It provides a simple and user-friendly interface that allows developers to write code efficiently. The Arduino IDE uses a programming language based on C/C++, which enables precise control over hardware operations.

The platform includes built-in libraries and supports external libraries that simplify tasks such as Bluetooth communication, sensor integration, and motor control. These libraries help reduce development time and effort. The Arduino IDE also provides tools like a serial monitor for debugging and monitoring real-time data.

Another important advantage of Arduino IDE is its open-source nature and strong community support. Developers can access documentation, tutorials, and sample codes, making development easier. In this project, the Arduino IDE plays a key role in integrating all system components and ensuring smooth communication between hardware and software.

3. Communication – Bluetooth

Bluetooth technology is used as the primary communication medium between the user's smartphone and the robot. It enables wireless data transfer over short distances without requiring an internet connection, making it highly suitable for real-time control applications. In this system, the user provides voice input through the mobile application, which is then processed and converted into a digital command signal.

This signal is transmitted via Bluetooth to the ESP32, where it is received and decoded. Based on the decoded command, the ESP32 triggers the corresponding action in the robot. The use of Bluetooth ensures low latency communication, which is essential for immediate response and smooth operation of the robot. It also consumes less power compared to other wireless technologies, thereby increasing battery efficiency.

Bluetooth communication is highly reliable and widely supported by most modern smartphones, making the system easily accessible and user-friendly. Additionally, it simplifies the overall system design since no additional networking infrastructure is required. This makes Bluetooth an ideal choice for implementing real-time wireless control in embedded robotic systems.

4. Motor Driver – L298N

The L298N motor driver module is a crucial component that acts as an interface between the ESP32 microcontroller and the DC motors of the robot. Since the ESP32 operates at low voltage levels and cannot supply sufficient current to drive motors directly, the motor driver is used to provide the necessary power amplification. It ensures that the motors receive adequate voltage and current to function effectively.

The L298N module is based on an H-bridge circuit configuration, which allows control over both the direction and speed of the motors. By changing the polarity of the voltage applied to the motors, the robot can move forward or backward. Additionally, by controlling individual motors, the robot can perform turning operations such as left and right movements.

The module also supports Pulse Width Modulation (PWM), which enables precise control over motor speed. This allows smooth acceleration, deceleration, and better maneuverability of the robot. Furthermore, the L298N includes built-in protection features such as heat sinks to dissipate heat generated during operation and voltage regulators to maintain stable performance. Its reliability and efficiency make it widely used in robotic and automation applications.

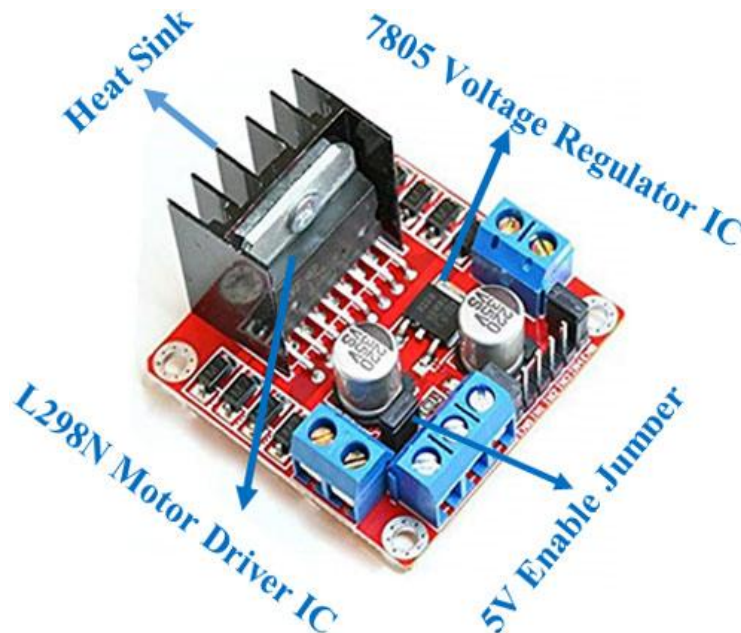


Figure 2.2: Motor Driver

5. Sensors – IR Sensor, Voltage Sensor

Sensors play a vital role in enhancing the intelligence, safety, and reliability of the robot. The Infrared (IR) sensor is used for obstacle detection and avoidance. It works by emitting infrared radiation and detecting the reflected signals from nearby objects. When an object is detected within a certain range, the sensor sends a signal to the ESP32, which then processes this information and takes appropriate action, such as stopping the robot or changing its direction. This feature helps prevent collisions and allows the robot to operate safely in dynamic environments.

The voltage sensor is used to monitor the battery level of the robot continuously. It measures the voltage of the power supply and sends this information to the ESP32 for analysis. If the voltage drops below a predefined threshold, the system can generate an alert or take preventive measures to avoid sudden shutdowns or damage to electronic components. This ensures uninterrupted operation and increases the overall lifespan of the system.

Together, these sensors provide essential feedback to the system, enabling it to make intelligent decisions and operate more efficiently. They enhance the robot's autonomy and improve user experience by ensuring safe and reliable performance.



Figure 2.3:IR Sensor

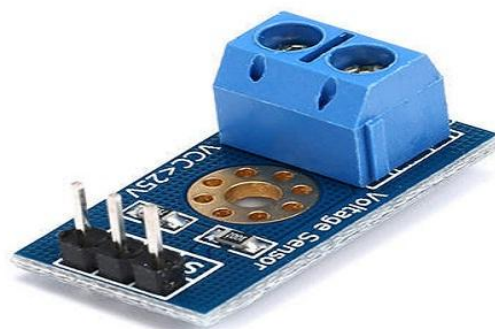


Figure 2.4:Voltage Sensor

6. Mobile Application – Arduino Bluetooth Control App

The Arduino Bluetooth Control App serves as the primary interface between the user and the robot, enabling intuitive and convenient operation. This application is installed on a smartphone and allows users to control the robot using voice commands. It utilizes the smartphone's built-in speech recognition technology to convert spoken words into text or predefined command signals.

Once the command is generated, it is transmitted to the ESP32 via Bluetooth communication. The ESP32 then processes the command and performs the corresponding action. The application provides a simple and interactive user interface, making it easy for users to operate the robot without requiring technical expertise. It also supports real-time communication, ensuring that commands are executed instantly without noticeable delay.

The use of a mobile application eliminates the need for physical controllers, enabling hands-free operation and increasing accessibility. It also allows for easy customization and future enhancements, such as adding new commands or integrating additional features. This makes the system more flexible, user-friendly, and suitable for real-world IoT and automation applications.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

In traditional robotic systems, robots are mainly controlled using wired connections, RF remote controls, or manual switches. These systems typically lack internet connectivity and cannot provide remote monitoring features. Voice control, if implemented, is often limited to short-range Bluetooth communication without cloud integration.

Most existing robots also do not include real-time feedback mechanisms. Users may not receive information about battery status, obstacle detection, or system errors. Additionally, many low-cost systems use basic microcontrollers without built-in IoT capabilities, which increases hardware complexity when adding wireless modules separately.

Disadvantages

- Limited control range due to wired or short-range communication.
- Lack of IoT integration and remote monitoring capabilities.
- No real-time feedback on robot status.
- Increased hardware complexity when adding external Wi-Fi modules.
- Poor obstacle detection and safety mechanisms in basic designs.
- Low flexibility and scalability for future enhancements.

3.2 PROBLEM STATEMENT

The current robotic systems are inefficient in providing smart control and monitoring features. Users cannot control robots conveniently using natural interfaces like voice, nor can they monitor system status remotely. There is a need to develop an intelligent robotic system that supports voice-based control, wireless communication, and real-time feedback using IoT technologies while maintaining low cost and simplicity.

3.3 PROPOSED SYSTEM

The proposed Voice-Controlled IoT Robot uses the ESP32 microcontroller as the central control unit to enable wireless voice command processing and IoT connectivity. Voice commands are transmitted through a smartphone application or cloud interface and processed by the ESP32 to control motor movement via a motor driver module.

IR sensors are incorporated for obstacle detection to ensure safe navigation. A voltage sensor monitors battery levels and sends alerts when power is low. The buzzer provides audible feedback for warnings and confirmations. Through IoT integration, real-time feedback such as movement status, obstacle alerts, and battery condition is sent back to the user.

This proposed system offers improved flexibility, long-range operation via the internet, enhanced safety features, and real-time monitoring. It is cost-effective, scalable, and suitable for smart automation applications in homes, industries, and research environments.

Advantages

- Hands-free operation using voice commands
- Real-time monitoring through IoT
- Improved safety with obstacle detection
- Low-cost and efficient system design
- Reduced hardware complexity (ESP32 has built-in Wi-Fi & Bluetooth)
- Scalable for future enhancements like AI and automation.

4. SYSTEM REQUIREMENTS SPECIFICATION

4.1 FUNCTIONAL REQUIREMENTS

The functional requirements define the core operations and functionalities that the Voice-Controlled IoT Robot must perform. These requirements ensure that the system behaves as expected and fulfills the intended objectives.

1. Accept and Process Voice Commands

The system must be capable of receiving voice commands from the user through a smartphone application. The mobile application converts the spoken commands into text or predefined signals and transmits them to the ESP32 via Bluetooth. The ESP32 should accurately interpret these commands without delay.

Supported commands include:

- Forward
- Backward
- Left
- Right
- Stop

The system should ensure proper recognition and avoid misinterpretation of commands.

2. Control Robot Movement in All Directions

The robot must be able to move in multiple directions based on the received commands. The ESP32 processes the input and sends control signals to the motor driver (L298N), which in turn drives the motors.

Movement functionalities include:

- Moving forward and backward
- Turning left and right
- Stopping instantly when required

The movement should be smooth, responsive, and accurately follow user commands.

3. Detect Obstacles

The system must include IR sensors to detect obstacles in the robot's path. When an obstacle is detected:

- The sensor sends a signal to the ESP32
- The system may stop or alert the user
- Prevents collisions and damage

This feature enhances safety and ensures reliable navigation in real-world environments.

4. Monitor Battery Voltage

The robot must continuously monitor its battery level using a voltage sensor. The ESP32 reads the voltage data and:

- Determines battery status (normal/low)
- Sends alerts when battery is low
- Prevents sudden shutdown

This ensures uninterrupted operation and helps in maintaining system efficiency.

5. Send Real-Time IoT Updates

The system must provide real-time updates to the user through an IoT platform or mobile interface. These updates include:

- Robot movement status
- Obstacle detection alerts
- Battery condition

The ESP32 transmits this data using Wi-Fi or Bluetooth, enabling remote monitoring and better user interaction.

6. Ensure Continuous System Operation

The system must operate continuously without interruptions. It should:

- Handle multiple commands efficiently
- Maintain stable communication
- Recover from minor errors automatically

4.2 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality, performance, and usability aspects of the system. These requirements ensure that the system operates efficiently, reliably, and provides a good user experience.

1. High Reliability

The system must operate consistently without failures under normal working conditions. The ESP32, sensors, and communication modules should function accurately even during continuous operation. The robot should respond correctly to voice commands without errors or delays. Any unexpected failures should be minimal, and the system should be able to recover quickly.

2. Energy Efficiency

The system should consume minimal power to ensure longer battery life. Efficient components such as the ESP32 and optimized code should be used to reduce energy consumption. Sensors and modules should operate only when required to avoid unnecessary power usage, making the robot suitable for prolonged use.

3. User-Friendly Interface

The mobile application used for voice control should be simple and easy to use. Users should be able to give commands without technical knowledge. The interface should clearly display system status, including movement, battery level, and alerts, ensuring a smooth and intuitive user experience.

4. Low Battery Usage

The system must optimize battery usage by minimizing unnecessary operations. Power-efficient components and proper voltage regulation should be used. The robot should also provide alerts when the battery level is low to prevent sudden shutdown and ensure continuous operation.

5. Performance Efficiency

The system should provide fast response to voice commands with minimal latency. Bluetooth communication and processing by ESP32 should be optimized to ensure real-time control without noticeable delays.

4.3 SOFTWARE REQUIREMENTS

The software requirements define the tools and platforms used for developing, controlling, and monitoring the Voice-Controlled IoT Robot system.

1. Arduino IDE : Used to write and upload code to ESP32
2. ESP32 Board Support Package (BSP) : Helps Arduino IDE support ESP32
3. Bluetooth Control Application : Used to send voice commands from mobile
4. IoT Cloud Platform : Used to monitor robot data online

4.4 HARDWARE REQUIREMENTS

The hardware components form the physical structure of the Voice-Controlled IoT Robot. Each component plays a crucial role in ensuring proper functioning, control, and monitoring of the system.

1. ESP32 Microcontroller : Main controller used to process commands and control the robot
2. L298N Motor Driver : Used to control motor speed and direction
3. DC Motors : Used for movement of the robot
4. IR Sensor : Used to detect obstacles
5. Voltage Sensor : Used to monitor battery level
6. Buzzer : Used to give alert sound
7. Battery & Power Supply : Provides power to all components

5. SYSTEM DESIGN

5.1 SYSTEM SPECIFICATIONS

The system specifications define the overall technical architecture, hardware components, communication methods, and operational characteristics of the Voice-Controlled IoT Robot. These specifications ensure that the system performs efficiently, reliably, and in real time while maintaining user convenience and safety.

1. Microcontroller: ESP32

The ESP32 microcontroller serves as the core processing unit (brain) of the entire system. It is responsible for executing all logical operations, processing user commands, and coordinating between input and output devices.

The ESP32 is chosen due to its high performance, low power consumption, and integrated communication features. It is a dual-core processor capable of handling multiple tasks simultaneously, making it suitable for real-time applications such as robotic control.

Additionally, the ESP32 comes with built-in Wi-Fi and Bluetooth modules, eliminating the need for external communication hardware. This reduces system complexity, cost, and power usage. It also supports GPIO pins for interfacing with sensors, motors, and other peripherals, ensuring flexible hardware integration.

2. Communication: Bluetooth and Wi-Fi

The system uses a hybrid communication approach combining both Bluetooth and Wi-Fi technologies to enhance functionality and flexibility.

Bluetooth is used for short-range, real-time communication between the smartphone and the robot. When a user gives a voice command through a mobile application, the command is transmitted instantly to the ESP32 via Bluetooth. This ensures low latency and quick response during robot operation.

Wi-Fi enables the IoT functionality of the system. It allows the robot to connect to the internet and send data such as battery status, obstacle detection alerts, and system health information to cloud platforms or mobile dashboards. This feature enables remote monitoring and future scalability.

3. Input System

The primary input to the system is voice commands provided by the user through a smartphone application.

When the user speaks a command (such as “forward”, “left”, or “stop”), the mobile application converts the speech into text using speech recognition technology. This text command is then converted into a digital signal and transmitted wirelessly via Bluetooth to the ESP32.

The ESP32 receives the command, interprets it, and triggers the appropriate action. This method of input eliminates the need for physical controllers or buttons, making the system more user-friendly and accessible, especially for hands-free operation.

4. Output System: Motor Movement and Alerts

The system generates outputs in two main forms: robot movement and alert signals.

Motor Movement:

Based on the received commands, the ESP32 controls the motors through a motor driver module. The robot can perform movements such as:

- Forward
- Backward
- Left turn
- Right turn
- Stop

These movements allow the robot to navigate in different directions as instructed by the user.

Alert System (Buzzer):

A buzzer is used to provide audio feedback and alerts. For example, it can beep when an obstacle is detected or when the battery level is low. This improves user awareness and enhances safety during operation.

5. Sensors: IR Sensor and Voltage Sensor

Sensors play a crucial role in enabling environmental awareness and system monitoring.

IR Sensor (Obstacle Detection):

The Infrared (IR) sensor detects obstacles in the path of the robot. When an object is detected, the sensor sends a signal to the ESP32, which can either stop the robot or alert the user. This helps prevent collisions and ensures safe navigation.

Voltage Sensor (Battery Monitoring):

The voltage sensor continuously monitors the battery level of the robot. It provides real-time feedback about the power status, allowing the system to notify the user when the battery is low. This helps in preventing sudden shutdowns and ensures efficient power management.

6. Control Type: Wireless and Voice-Based

The system operates entirely on a wireless and voice-controlled mechanism, eliminating the need for manual controls such as switches or joysticks.

Users interact with the robot through voice commands, making the system intuitive and easy to operate. This is particularly useful in scenarios where hands-free control is required.

The wireless nature of the system ensures flexibility, mobility, and ease of use, allowing the robot to be controlled from a distance without physical connections. This also makes the system scalable for future enhancements such as remote internet-based control.

5.2 SYSTEM COMPONENTS

The system is divided into multiple modules, each responsible for a specific function. This modular design improves flexibility, maintainability, and scalability.

1. Voice Input Module

This module is responsible for capturing voice commands from the user. A smartphone application is used to receive spoken instructions and convert them into predefined command signals. These signals correspond to robot actions such as movement directions. The module ensures that user interaction is simple, intuitive, and hands-free.

2. Communication Module

The communication module handles the wireless transfer of data between the smartphone and the robot. Bluetooth technology is used for transmitting commands from the mobile application to the ESP32. This module ensures fast, reliable, and low-latency communication. Additionally, Wi-Fi can be used for sending feedback data to IoT platforms for remote monitoring.

3. Processing Module

The processing module consists of the ESP32 microcontroller, which acts as the brain of the system. It receives commands from the communication module, interprets them, and makes decisions accordingly. It also processes data from sensors and determines appropriate actions such as stopping the robot when an obstacle is detected or triggering alerts.

4. Motor Control Module

This module includes the L298N motor driver and DC motors. The ESP32 sends control signals to the motor driver, which regulates the power supplied to the motors. Based on these signals, the motors rotate in specific directions, enabling the robot to move forward, backward, left, or right. The module ensures smooth and precise movement.

5. Sensor Module

The sensor module consists of IR sensors and a voltage sensor.

- IR Sensor: Detects obstacles by emitting infrared rays and sensing reflections. It helps in avoiding collisions.
- Voltage Sensor: Monitors the battery level and provides data to the ESP32 for analysis.

This module enhances safety and ensures efficient operation of the robot.

6. Alert Module

The alert module includes a buzzer that provides audio signals to notify the user about system conditions. It is activated when:

- An obstacle is detected
- Battery level is low
- Any abnormal condition occurs.

5.3 UML DIAGRAMS

Activity Diagram

An activity diagram shows the flow of activities in a system from start to end. It represents processes, decisions, loops, and parallel actions. It helps in understanding how a system performs tasks step by step. It uses symbols like arrows, decision nodes, and start/end points. Commonly used for modeling workflows and business processes. It is useful during system design and analysis.

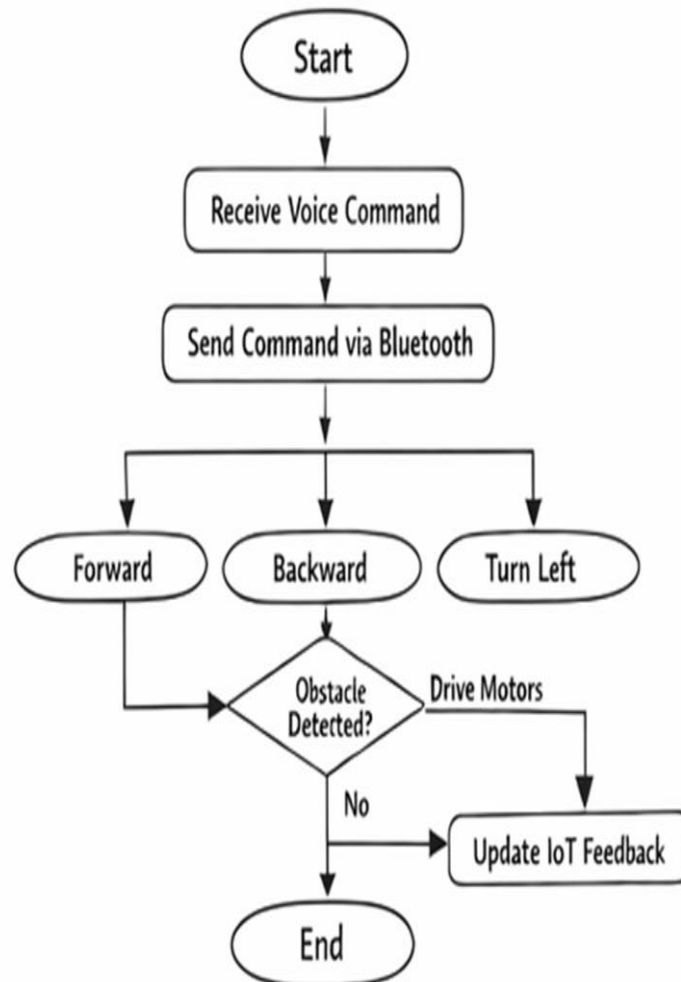


Figure 5.1: Activity Diagram

Usecase Diagram

A use case diagram represents the interaction between users (actors) and the system. It shows what functions the system provides to the users. Each function is called a use case. It helps in capturing and visualizing system requirements clearly. Relationships like include, extend, and association are used. It is mainly used in the initial stages of software development.

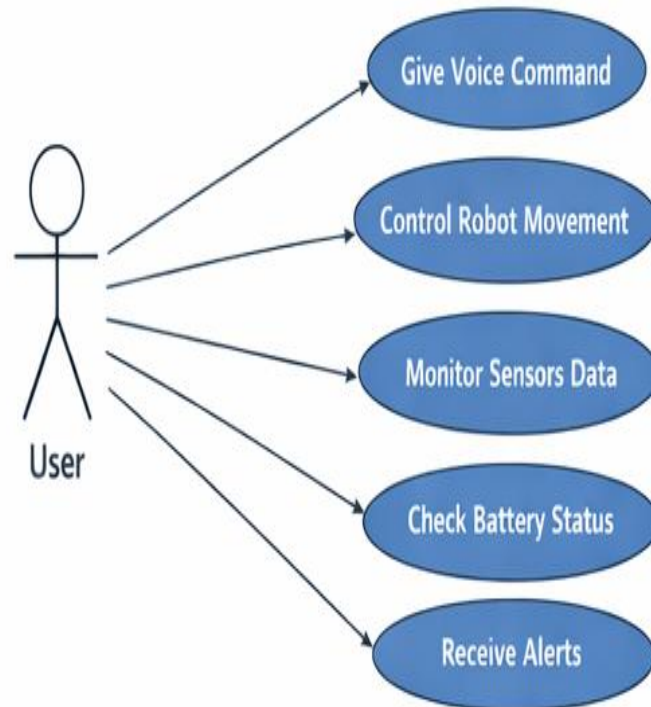


Figure 5.2: Usecase Diagram

Class Diagram

A class diagram shows the structure of a system by representing classes. Each class contains attributes (data) and methods (functions). It also shows relationships like inheritance, association, and aggregation. It is a key diagram in object-oriented design. Helps developers understand system architecture. It is used during design and implementation phases.

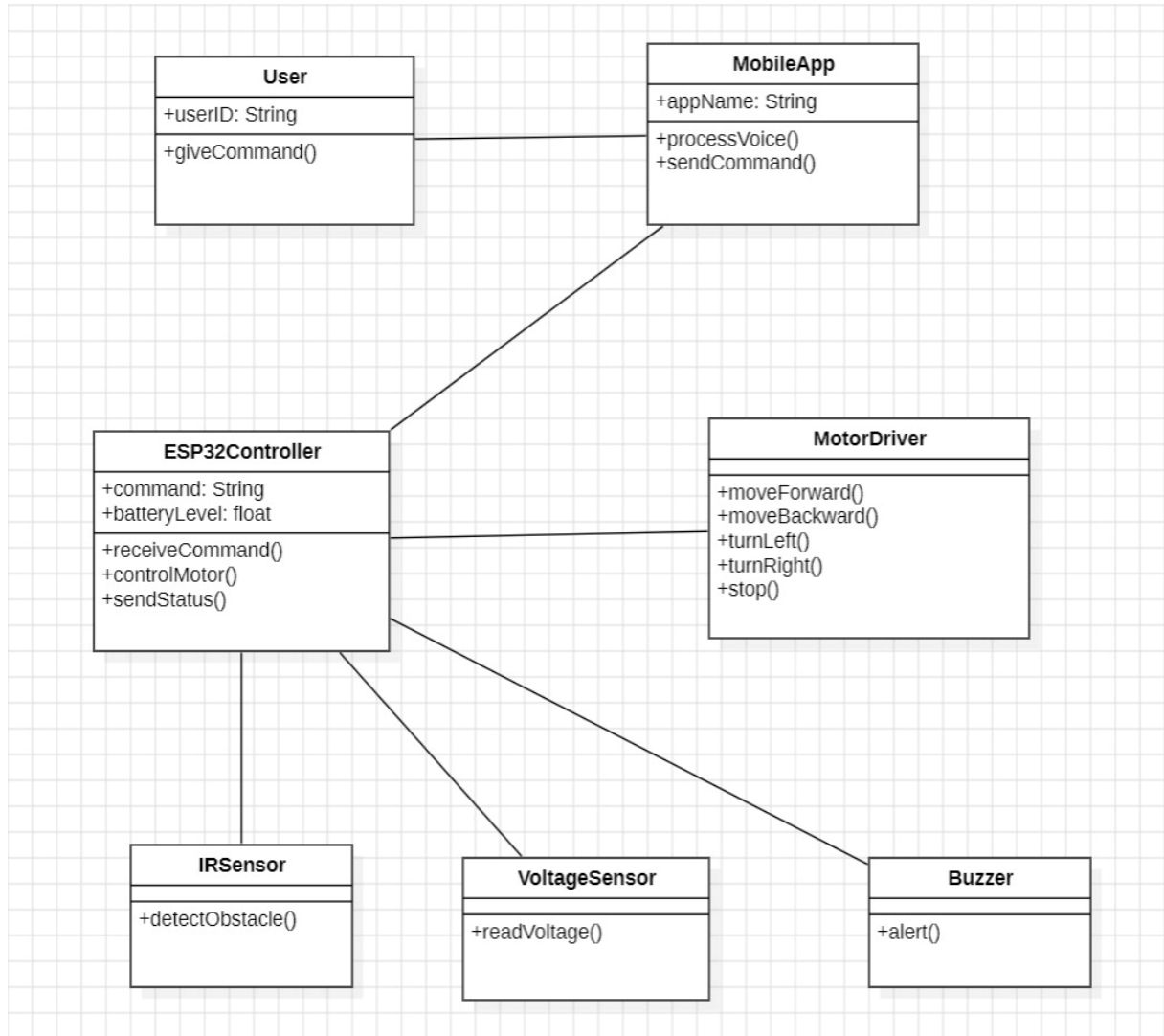


Figure 5.3: Class Diagram

Sequence Diagram

A sequence diagram shows how objects interact in a particular sequence. It focuses on the order of messages exchanged between objects. Time flows from top to bottom in the diagram. It helps in understanding the dynamic behavior of a system. Used to model real-time scenarios and use case flows. Useful for debugging and system design clarity.

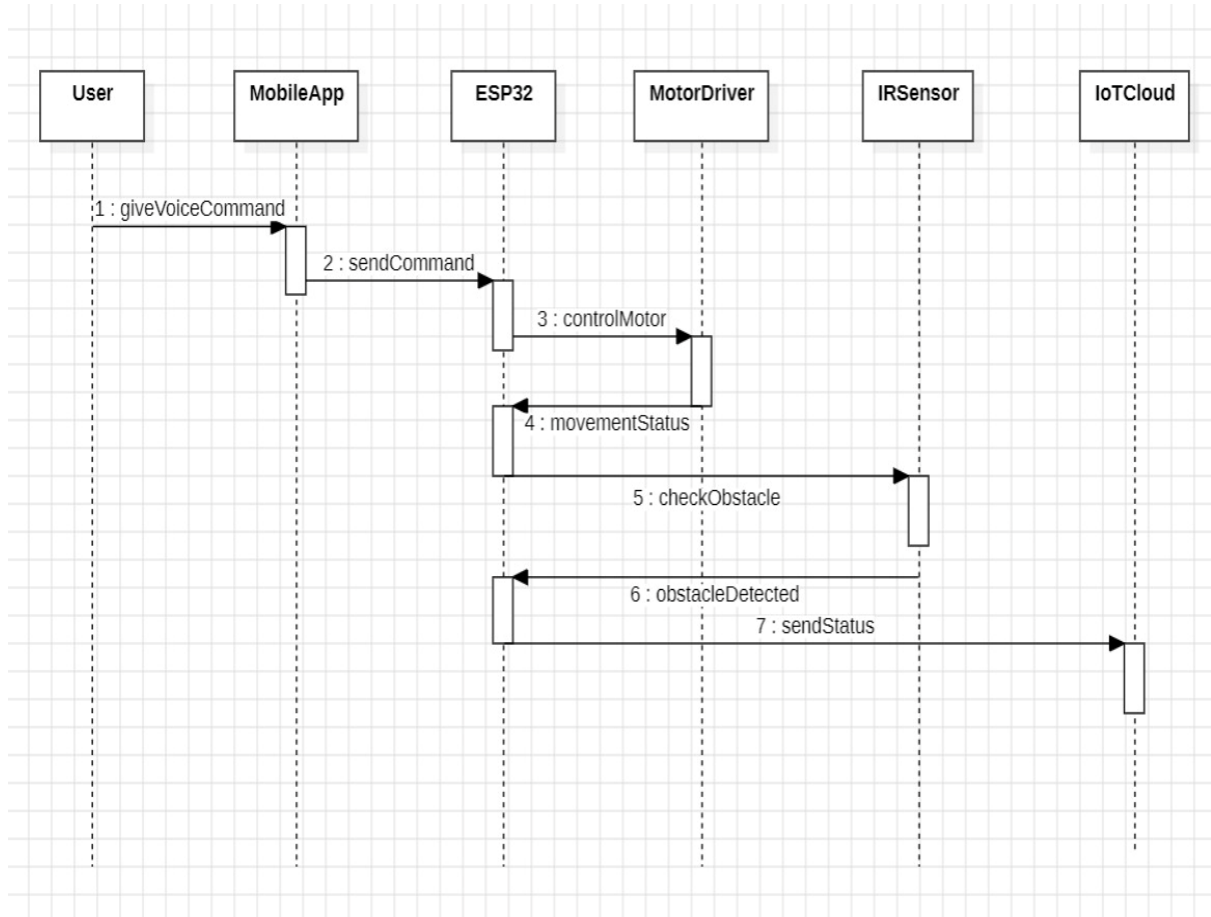


Figure 5.4: Sequence Diagram

6. IMPLEMENTATION

The ESP32 receives commands via Bluetooth and interprets them to control motors using the motor driver. Sensor data is continuously monitored and alerts are triggered when necessary.

6.1 SAMPLE CODE

```
#include<BluetoothSerial.h>

BluetoothSerial bt;

String data=" ";
String status=" ";

#define m1 33
#define m2 32
#define m3 25
#define m4 26

#define ANALOG_IN_PIN 35

float adc_voltage = 0.0;
float in_voltage = 0.0;
float R1 = 30000.0;
float R2 = 7500.0;
float ref_voltage = 3.3;
int adc_value = 0;

#define ir 27
int ir_sense;

#define buzzer 12
#define led 14
int i=0,j=0,k=0,l=0,m=0;

void setup()
{
  Serial.begin(9600);
  bt.begin("ESP VOICE ROBOT");
  pinMode(ANALOG_IN_PIN,INPUT);
  pinMode(ir,INPUT);
  pinMode(2,OUTPUT);
  digitalWrite(2,LOW);
  pinMode(buzzer,OUTPUT);
  digitalWrite(buzzer,LOW);
  pinMode(led,OUTPUT);
  digitalWrite(led,LOW);
```

```

pinMode(m1,OUTPUT);
pinMode(m2,OUTPUT);
pinMode(m3,OUTPUT);
pinMode(m4,OUTPUT);
digitalWrite(m1,LOW);
digitalWrite(m2,LOW);
digitalWrite(m3,LOW);
digitalWrite(m4,LOW);
}
void loop()
{
  ir_sense=digitalRead(ir);
  ir_sense=1-ir_sense;
  if(ir_sense==1)
  {
    data=" ";
    bt.println("OBJECT DETECTED");
    digitalWrite(buzzer,HIGH);
    stop();
    backward();
    delay(1000);
    stop();
  }
  else
  {
    digitalWrite(buzzer,LOW);
    bt_loop();
  }
  volt_loop();
  digitalWrite(2,HIGH);delay(50);digitalWrite(2,LOW);delay(50);
}
void bt_loop()
{
  if(bt.available()>0)
  {
    data=" ";

```

```

data=bt.readString();
Serial.print("data : ");Serial.println(data);
}
if(data=="FORWARD" || data=="Forward" || data=="forward")
{
    forward();
}
else if(data=="BACKWARD" || data=="Backward" || data=="backward")
{
    backward();
}
else if(data=="LEFT" || data=="Left" || data=="left")
{
    left();
    delay(100);
    stop();
    delay(100);
}
else if(data=="RIGHT" || data=="Right" || data=="right")
{
    right();
    delay(100);
    stop();
    delay(100);
}
else if(data=="STOP" || data=="Stop" || data=="stop")
{
    stop();
}
else if(data=="LIGHT" || data=="Light" || data=="light")
{
    data=" ";
    i=i+1;
    if(i==1)
    {
        digitalWrite(led,HIGH);
    }
}

```

```

    }
    else
    {
        i=0;
        digitalWrite(led,LOW);
    }
}
else if(data=="SOUND" || data=="Sound" || data=="sound")
{
    data=" ";
    j=j+1;
    if(j==1)
    {
        digitalWrite(buzzer,HIGH);
        delay(500);
    }
    else
    {
        j=0;
        digitalWrite(buzzer,LOW);
    }
}
else if(data=="VOLTAGE" || data=="Voltage" || data=="voltage")
{
    data=" ";
    String val=String("VOLTAGE :")+String(in_voltage)+String("V");
    bt.println(val);
}
}
void forward()
{
    digitalWrite(m1,HIGH);
    digitalWrite(m2,LOW);
    digitalWrite(m3,HIGH);
    digitalWrite(m4,LOW);
}

```

```

void backward()
{
    digitalWrite(m1,LOW);
    digitalWrite(m2,HIGH);
    digitalWrite(m3,LOW);
    digitalWrite(m4,HIGH);
}

void right()
{
    digitalWrite(m1,HIGH);
    digitalWrite(m2,LOW);
    digitalWrite(m3,LOW);
    digitalWrite(m4,HIGH);
}

void left()
{
    digitalWrite(m1,LOW);
    digitalWrite(m2,HIGH);
    digitalWrite(m3,HIGH);
    digitalWrite(m4,LOW);
}

void stop()
{
    digitalWrite(m1,LOW);
    digitalWrite(m2,LOW);
    digitalWrite(m3,LOW);
    digitalWrite(m4,LOW);
}

void volt_loop()
{
    adc_value = analogRead(ANALOG_IN_PIN);
    adc_voltage = (adc_value * ref_voltage) / 4096.0;
    in_voltage = adc_voltage*(R1+R2)/R2;
    Serial.print("Input Voltage = ");
    Serial.println(in_voltage, 2);
    delay(50);
}

```

```
if(in_voltage >= 6 || in_voltage <= 3)
{
    digitalWrite(buzzer,HIGH);
    bt.println("VOLTAGE FLUCTUATION DETECTED");
}
else
{
    digitalWrite(buzzer,LOW);
}
}
```

7. SYSTEM TESTING

7.1 TYPES OF TESTING

- **Unit Testing** : Testing individual modules separately to ensure each component functions correctly without errors.
- **Integration Testing** : Testing the interaction between combined modules to verify proper data flow and coordination.
- **System Testing** : Testing the entire system as a whole to ensure it meets all functional requirements and works properly.

7.2 TEST CASES

Table 7.1: Forward Movement Verification

Testcase#1	Priority (H, L): High	
Test Objective	To verify forward movement of the robot.	
Test Description	Robot should move forward when a voice command is given.	
Requirements verified	Bluetooth communication and motor control.	
Test Environment	Mobile application, ESP32, Robot	
Actions	Expected	Actual
Forward command	Robot will move forward	Robot moved forward
Pass: Yes	Conditional pass: No	Fail: No

Table 7.2: Backward Movement Verification

Testcase#2	Priority (H, L): High	
Test Objective	To verify backward movement.	
Test Description	Robot should move backward on command.	
Requirements verified	Motor control functionality.	
Test Environment	Full system	
Actions	Expected	Actual
Backward command	Robot will move backward	Robot moved backward
Pass: Yes	Conditional pass: No	Fail: No

Table 7.3 Left Direction Control

Testcase#3	Priority (H, L): High	
Test Objective	To verify left turn.	
Test Description	Robot should turn left when command is given.	
Requirements verified	Motor direction control.	
Test Environment	Full system	
Actions	Expected	Actual
Left command	Robot turns left	Robot turned left
Pass: Yes	Conditional pass: No	Fail: No

Table 7.4: Right Direction Control

Testcase#4	Priority (H, L): High	
Test Objective	To verify right turn.	
Test Description	Robot should turn right when command is given.	
Requirements verified	Motor direction control.	
Test Environment	Full system	
Actions	Expected	Actual
Right command	Robot turns right	Robot turned right
Pass: Yes	Conditional pass: No	Fail: No

Table 7.5: Stop Function Verification

Testcase#5	Priority (H, L): High	
Test Objective	To verify stop functionality.	
Test Description	Robot should stop immediately on command.	
Requirements verified	Safety and motor control.	
Test Environment	Full system	
Actions	Expected	Actual
Stop command	Robot stops immediately	Robot stoped immediately
Pass: Yes	Conditional pass: No	Fail: No

Table 7.6: Obstacle Detection using IR Sensor

Testcase#6	Priority (H, L): High	
Test Objective	To verify obstacle detection.	
Test Description	Robot should detect obstacle using IR sensor.	
Requirements verified	IR sensor functionality.	
Test Environment	Robot with IR sensor	
Actions	Expected	Actual
Obstacle placed	Robot stops and alerts	Robot stops and alerts
Pass: Yes	Conditional pass: No	Fail: No

Table 7.7: Battery Voltage Monitoring

Testcase#7	Priority (H, L): High	
Test Objective	To verify battery monitoring.	
Test Description	System should detect low voltage.	
Requirements verified	Voltage sensor.	
Test Environment	ESP32 system	
Actions	Expected	Actual
Low voltage	Alert generates	Alert generated
Pass: Yes	Conditional pass: Yes	Fail: No

Table 7.8: Bluetooth Connectivity Check

Testcase#8	Priority (H, L): High	
Test Objective	To verify Bluetooth connectivity.	
Test Description	System should connect to mobile device.	
Requirements verified	Bluetooth communication	
Test Environment	Mobile + ESP32	
Actions	Expected	Actual
Connect Bluetooth	Connection successful	Connection established
Pass: Yes	Conditional pass: No	Fail: No

7.3 DISCUSSION OF RESULTS

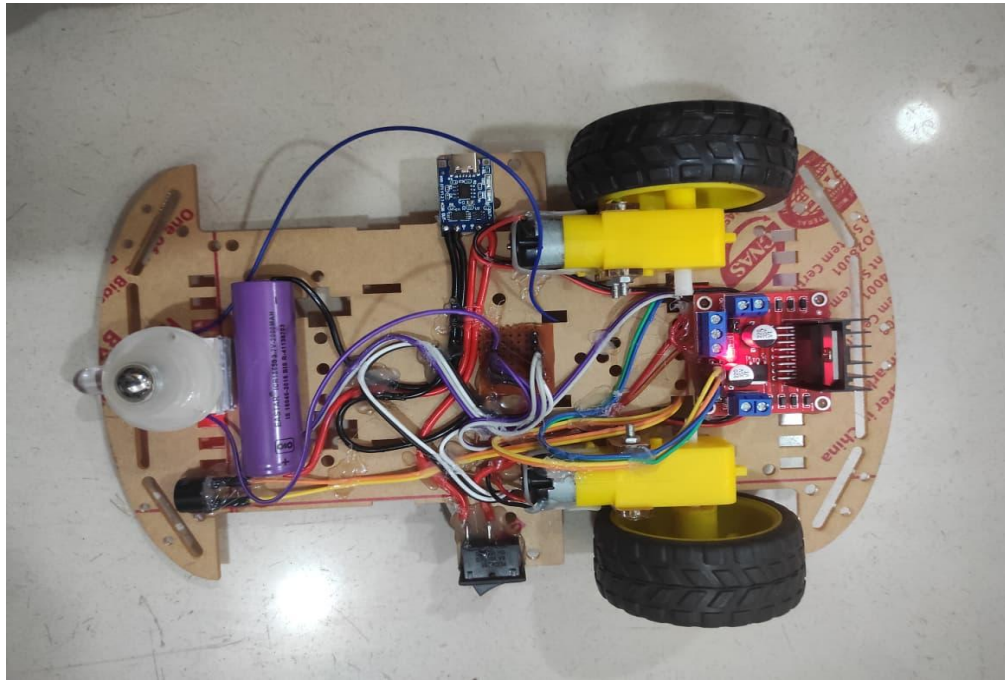


Figure 7.1. Underside of the robot car

- The image shows the underside of a robot car chassis with its core electronic setup.
- A lithium-ion battery (purple cell) is used as the main power source.
- A motor driver module (red board, likely L298N) controls the motors.
- A TP4056 charging module (blue board) is included for battery charging.
- Two DC gear motors with yellow casings drive the rear wheels.
- A caster wheel is placed at the front for balance and smooth movement.
- A toggle switch is provided for power control (ON/OFF).
- All components are mounted on an MDF/wooden chassis.
- The wiring connects all modules in a compact and organized layout.

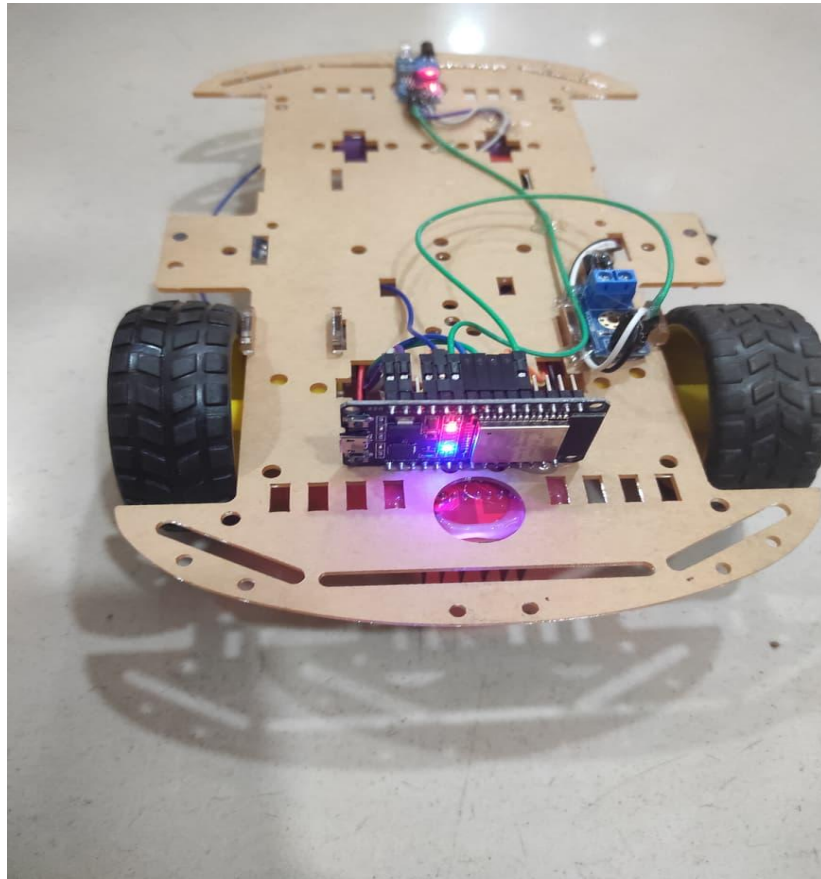
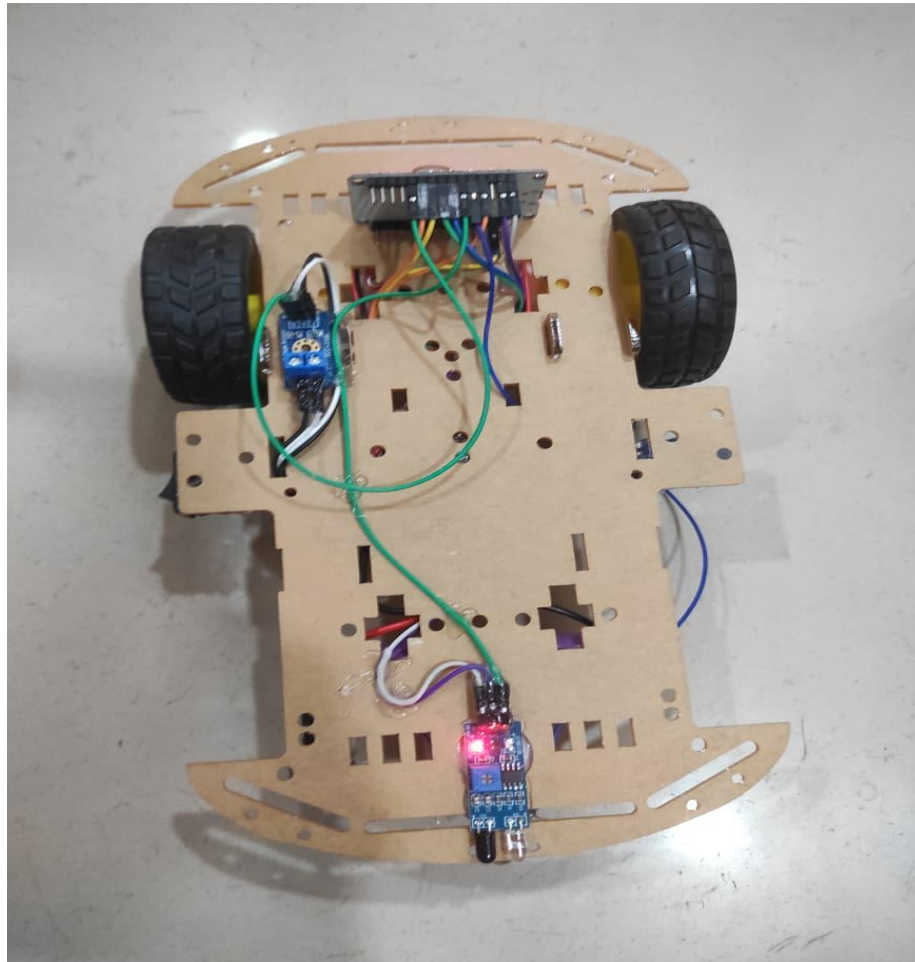


Figure 7.2: ESP8266 NodeMCU microcontroller

- The image shows the **top deck of the robot chassis**.
- An ESP8266 NodeMCU microcontroller is mounted at the center, acting as the brain of the system.
- It enables Wi-Fi connectivity for receiving voice commands.
- A glowing RGB/IR sensor is visible at the front-bottom, emitting a purple/pink light.
- An adjustable sensor module is mounted on the right side.
- Green jumper wires are used to connect all the components.
- The robot appears to be in an active and powered-on state.



7.3. IR obstacle detection sensor

- The image shows the bottom face of the robot chassis.
- An IR obstacle detection sensor (blue module with red LED) is mounted at the front-bottom.
- The sensor is used for detecting obstacles at ground level.
- A small trim potentiometer module is present for adjusting sensor sensitivity.
- Motor wiring is neatly routed towards the rear drive wheels.
- The glowing red LED indicates that the IR sensor is powered and actively detecting objects.



Figure 7.4: ESP Voice Robot mobile application

- The screenshot shows the **"ESP Voice Robot" mobile application** interface.
- The app is receiving real-time feedback from the robot over Wi-Fi.
- A chat-style interface displays repeated **"OBJECT DETECTED"** messages.
- These messages confirm that the IR sensor successfully detects obstacles.
- The alerts are transmitted to the app via the ESP8266 module.
- A command input box is available at the bottom of the screen.
- The user can send voice or text commands to the robot through this input.
- This demonstrates bidirectional IoT communication between the robot and the smartphone.

8. CONCLUSION AND FUTURE ENHANCEMENTS

8.1. CONCLUSION

The Voice-Controlled Robot with IoT Feedback presents a modern approach to robotic automation by integrating voice recognition, Bluetooth communication, and IoT monitoring using the ESP32. The system eliminates manual control limitations and enhances accessibility through hands-free operation. Real-time IoT feedback improves transparency, reliability, and monitoring efficiency.

The proposed system is cost-effective, scalable, and suitable for applications in smart homes, industrial automation, surveillance, and educational robotics. Future enhancements may include AI-based voice processing, camera integration, and cloud-based analytics for intelligent decision-making.

8.2. FUTURE ENHANCEMENTS

The current system can be further improved by incorporating advanced technologies to enhance functionality, intelligence, and user interaction. The following enhancements can be considered for future development:

Integration of Wi-Fi-Based Remote Access

The system can be upgraded to support full Wi-Fi control, allowing users to operate the robot from anywhere through the internet instead of being limited to Bluetooth range.

Addition of Camera Module for Surveillance

A camera (such as ESP32-CAM) can be integrated to enable live video streaming. This will make the robot suitable for surveillance, security monitoring, and remote inspection applications.

Human-Like Voice Interaction (Speaking Capability)

The system can be upgraded with Text-to-Speech (TTS) technology to allow the robot to respond verbally to user commands. This would enable interactive communication, where the robot can provide responses such as status updates, alerts, or confirmations in a human-like voice, making the system more intelligent and user-friendly.

Implementation of AI-Based Voice Recognition

Advanced voice recognition using Artificial Intelligence can be implemented to improve accuracy and allow the robot to understand natural language commands instead of predefined keywords.

Use of Advanced Sensors (Ultrasonic, GPS)

Ultrasonic sensors can provide more accurate obstacle detection, while GPS modules can enable location tracking and navigation, making the robot suitable for outdoor applications.

Development of a Custom Mobile Application

A dedicated mobile application can be developed with an improved user interface, real-time data visualization, and better control features compared to existing generic apps.

Autonomous Navigation Capability

The robot can be enhanced with autonomous navigation, enabling it to move independently without user commands using path planning and obstacle avoidance algorithms.

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